National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task: Signals

Lab Task 1: Handling Multiple Signals with `pause()`

Write a **program that installs signal handlers for SIGINT, SIGTERM, and SIGALRM**. The program should print a message when it receives these signals and continue running using `pause()`.

Instructions:

1. **Register signal handlers** for:
   - SIGINT (Ctrl+C) → Print "SIGINT received! Ignoring Ctrl+C."
   - SIGTERM → Print "SIGTERM received! Exiting program." and **exit immediately**.
   - SIGALRM → Print "SIGALRM received! Timer expired."
2. **Set an alarm (`alarm(10)`)** to trigger SIGALRM after **10 seconds**.
3. Use `pause()` to **keep the program running and waiting for signals**.

Lab Task 2: Parent-Child Signal Communication Using SIGQUIT

Objective:
Modify the parent-child communication program so that:

- The child process waits for 5 seconds and then sends **SIGQUIT** to the parent process.

- The parent process handles **SIGQUIT** and prints a message before terminating.

Instructions:

Parent Process:

- Register a signal handler for **SIGQUIT**.
- Wait for **SIGQUIT** using **pause()**.
- When **SIGQUIT** is received, print "Parent received SIGQUIT from child! Exiting..." and terminate the parent process.

Child Process:

- Sleep for 5 seconds.
- Send **SIGQUIT** to the parent using `kill(getppid(), SIGQUIT);`.
- Print "Child sent SIGQUIT to parent!" before exiting.

How to Test?

1. Run the program.
2. The child process waits for 5 seconds, then sends **SIGQUIT** to the parent.
3. The parent receives **SIGQUIT**, prints a message, and exits

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Lab Task 3: Process Synchronization Using `sigsuspend()` and `sigprocmask()`

Write a **program** where a **parent process waits for SIGUSR1 from a child** before continuing execution.

1. **Parent Process:**
    - **Block SIGUSR1** using `sigprocmask()`.
    - **Use `sigsuspend()` to wait for SIGUSR1**.

    - When SIGUSR1 is received, print "Received SIGUSR1, resuming execution.".
2. **Child Process:**
    - **Sleep for 5 seconds** and then **send SIGUSR1 to the parent**.
    - After sending the signal, **terminate gracefully**.
3. **Parent should detect child's termination using wait()**.

Instructions for Submission

- Ensure your code is **well-documented with comments**.
- Implement **proper error handling** Test your program with **various inputs** before submission.
- Submit your **source code along with screenshots of successful runs**.

THE END
